

# Complexity Analysis

Given graph:  $V = 6$  vertices and  $E = 6$  edges.

## 1. Adjacency Matrix

**Space:**  $O(V^2)$ . For this graph, the matrix contains  $6 \times 6 = 36$  entries.

**BFS:**  $O(V^2)$ . The execution required 36 neighbor checks.

**DFS:**  $O(V^2)$ . The execution required 36 neighbor checks.

**Edge checking:**  $O(1)$ , because  $\text{matrix}[u][v]$  can be accessed directly. The E-F check required 1 operation.

## 2. Adjacency List

**Space:**  $O(V + E)$ . It stores the vertices and their actual connections.

**BFS:**  $O(V + E)$ . The execution required 12 neighbor checks.

**DFS:**  $O(V + E)$ . The execution required 12 neighbor checks.

**Edge checking:**  $O(\text{degree of vertex})$ . For  $E \rightarrow B \rightarrow F$ , checking E-F required 2 neighbor comparisons.

## 3. Complexity Summary

| Operation     | Adjacency Matrix | Adjacency List     |
|---------------|------------------|--------------------|
| Space         | $O(V^2)$         | $O(V + E)$         |
| BFS           | $O(V^2)$         | $O(V + E)$         |
| DFS           | $O(V^2)$         | $O(V + E)$         |
| Edge checking | $O(1)$           | $O(\text{degree})$ |
| BFS checks    | 36               | 12                 |
| DFS checks    | 36               | 12                 |
| E-F check     | 1                | 2                  |